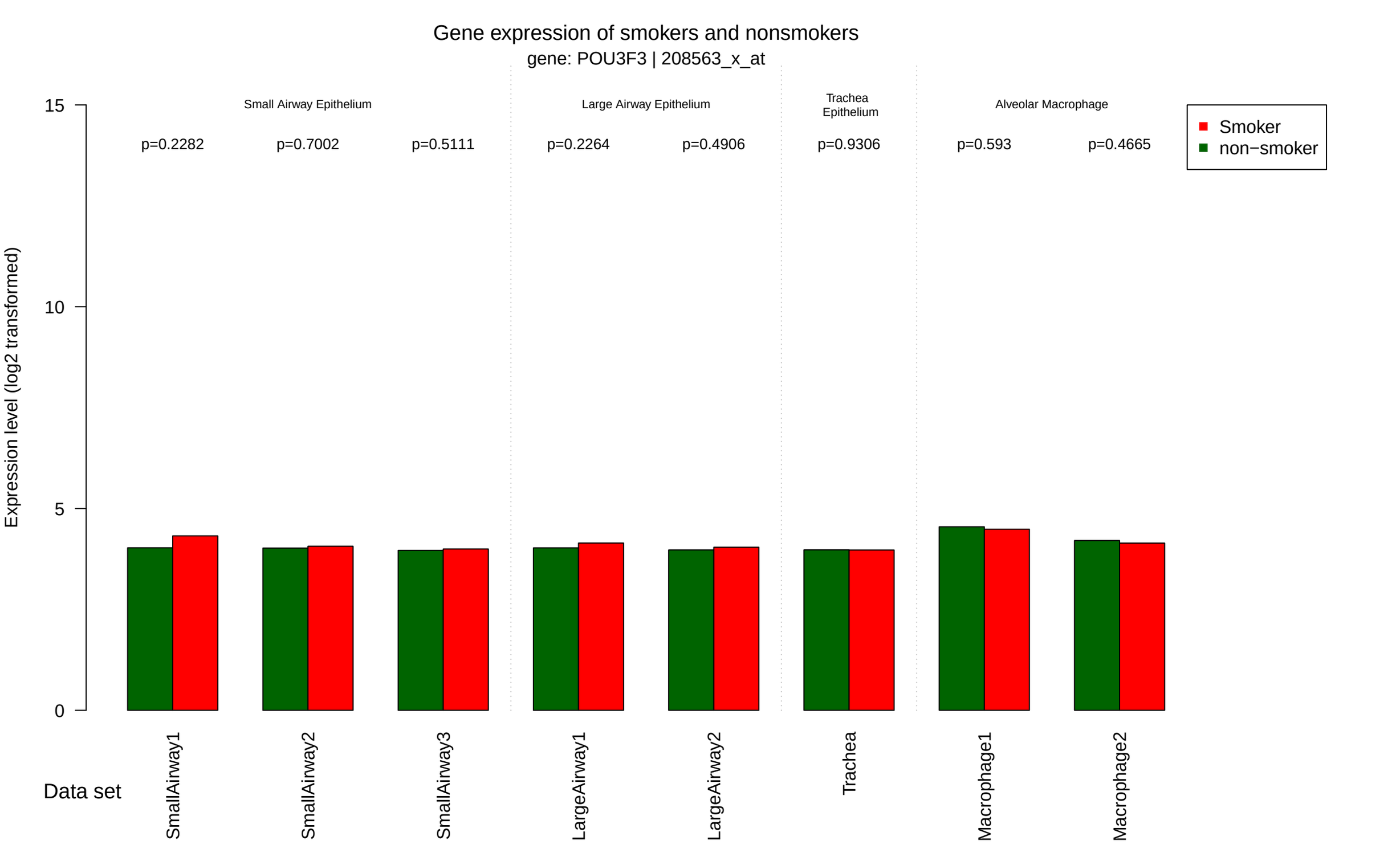
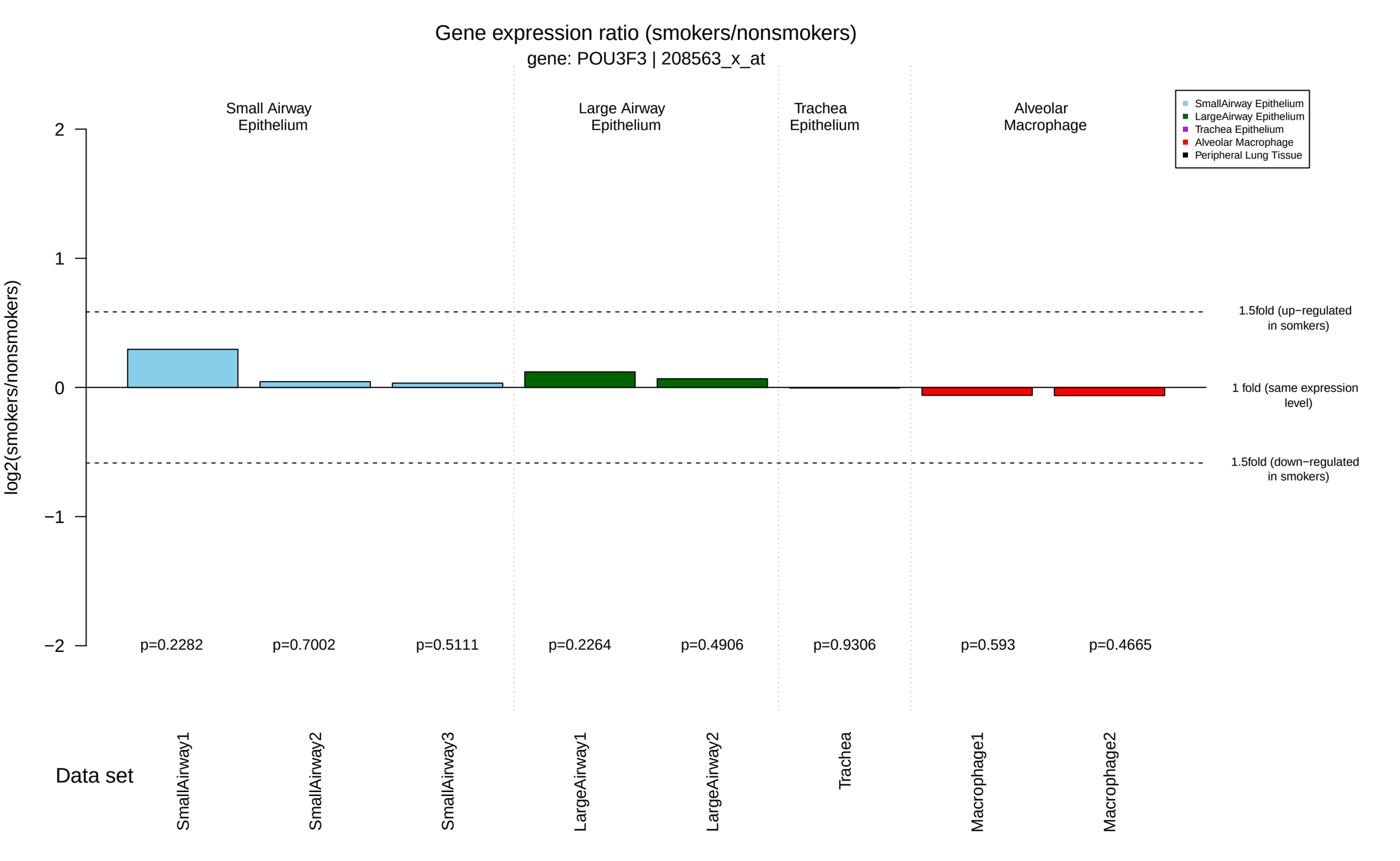
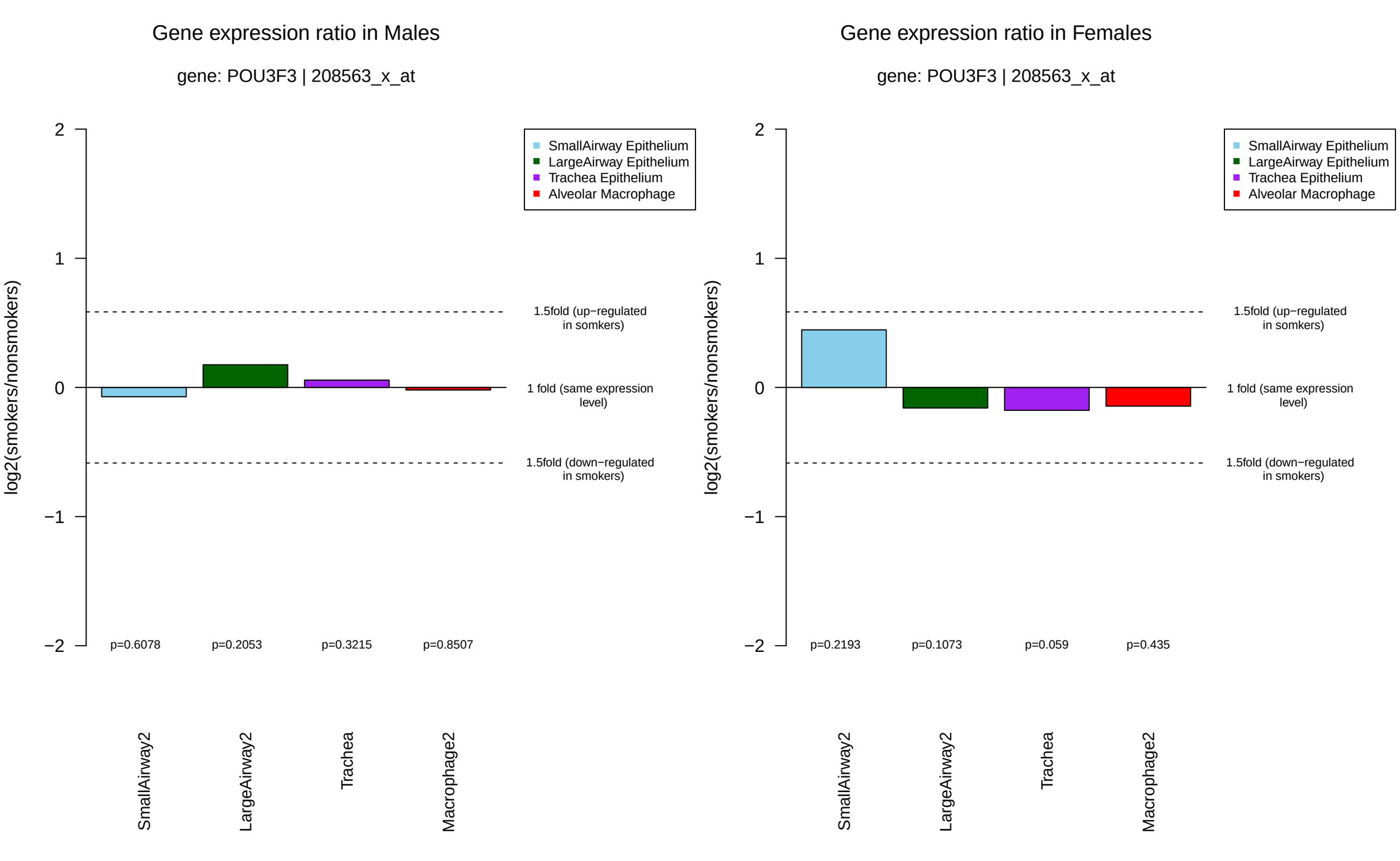
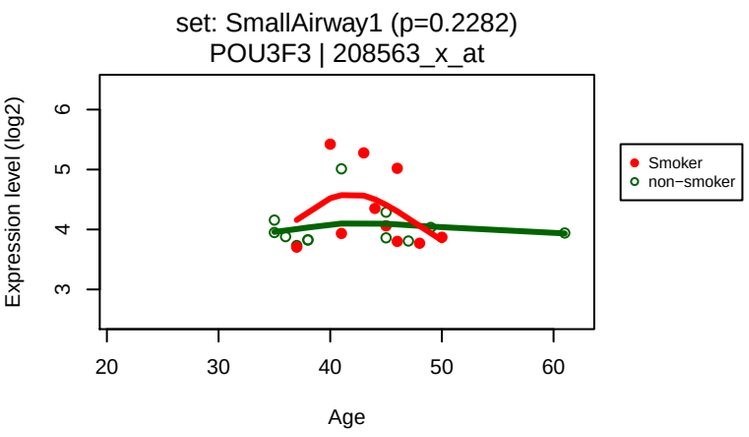
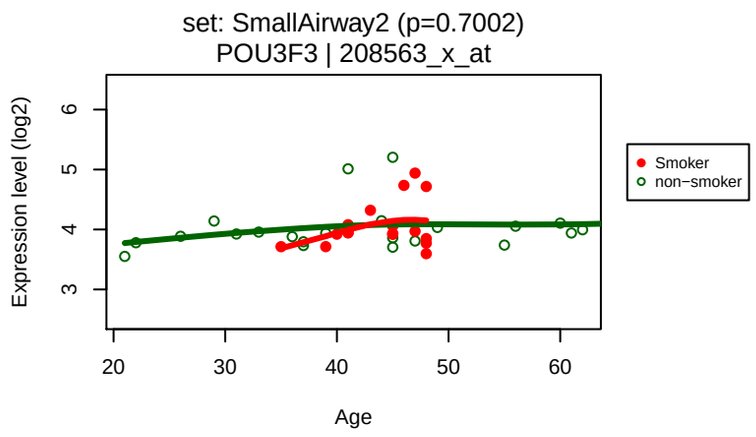
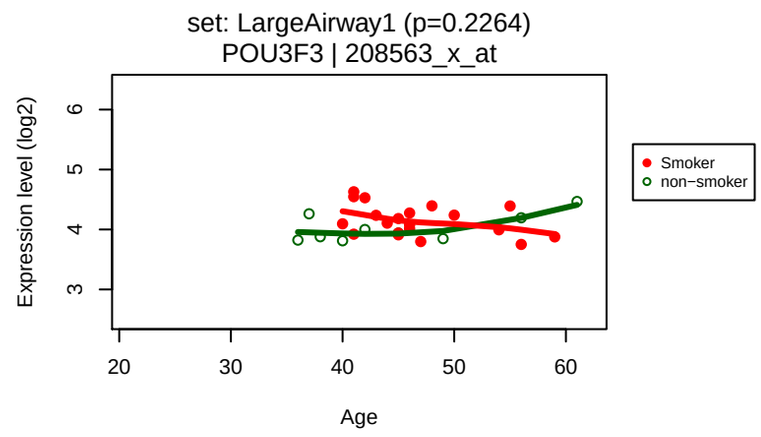

Gene expression of smokers and nonsmokers
gene: POU3F3 | 208563_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2282
p=0.7002
p=0.5111
p=0.2264
p=0.4906
p=0.9306
p=0.593
p=0.4665
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

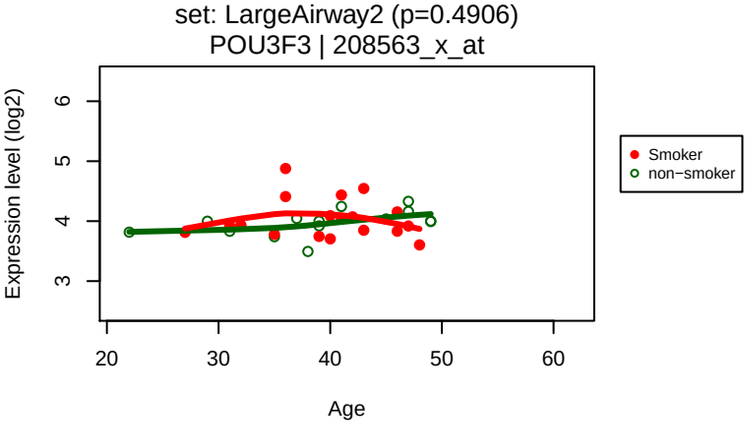

Gene expression ratio (smokers/nonsmokers)
gene: POU3F3 | 208563_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2282
p=0.7002
p=0.5111
p=0.2264
p=0.4906
p=0.9306
p=0.593
p=0.4665
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

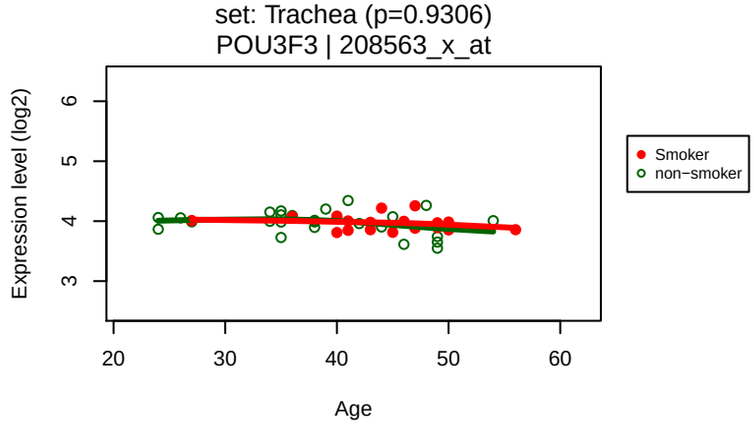

Gene expression ratio in Males
gene: POU3F3 | 208563_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6078
p=0.2053
p=0.3215
p=0.8507
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POU3F3 | 208563_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2193
p=0.1073
p=0.059
p=0.435
SmallAirway2
LargeAirway2
Trachea
Macrophage2

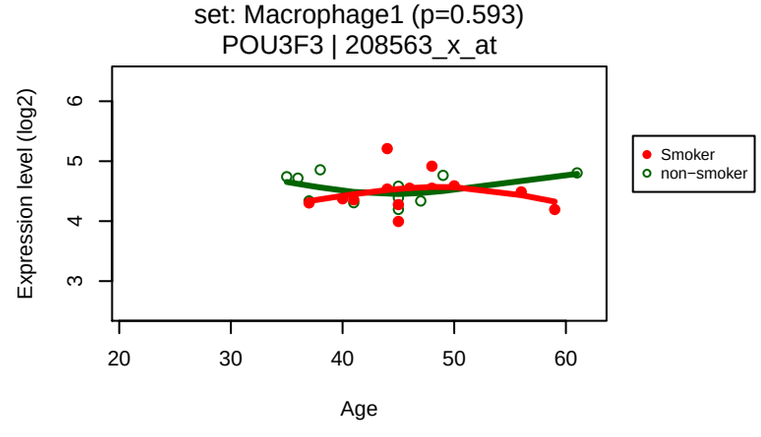

set: SmallAirway1 (p=0.2282)
POU3F3 | 208563_x_at
Expression level (log2)
Age
Smoker
non−smoker

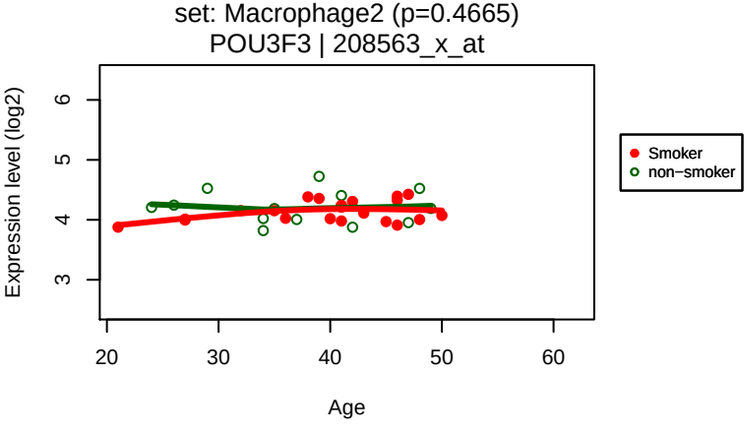

set: SmallAirway2 (p=0.7002)
POU3F3 | 208563_x_at
Expression level (log2)
Age
Smoker
non−smoker

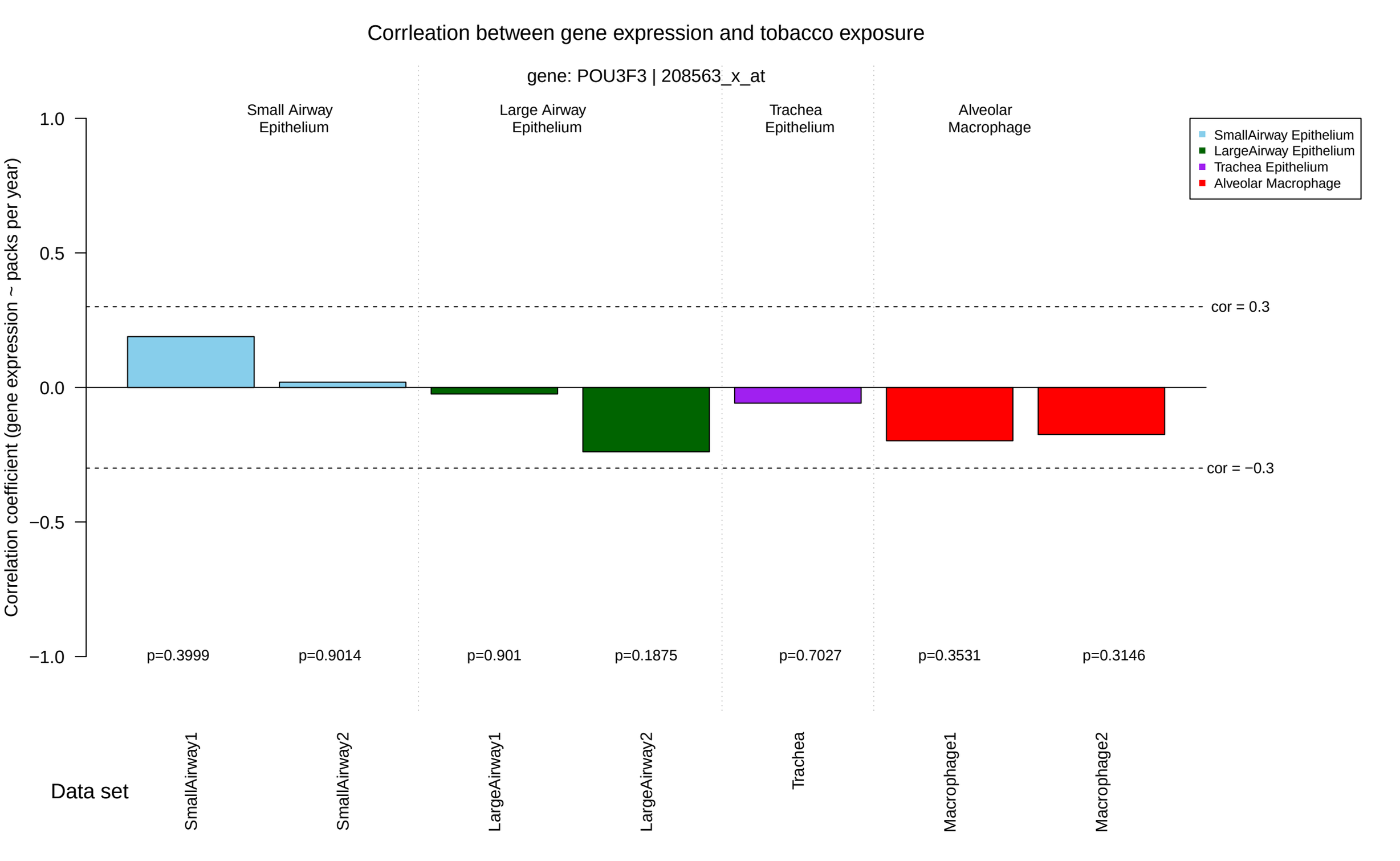

set: LargeAirway1 (p=0.2264)
POU3F3 | 208563_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4906)
POU3F3 | 208563_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9306)
POU3F3 | 208563_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.593)
POU3F3 | 208563_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4665)
POU3F3 | 208563_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POU3F3 | 208563_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3999
p=0.9014
p=0.901
p=0.1875
p=0.7027
p=0.3531
p=0.3146
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2